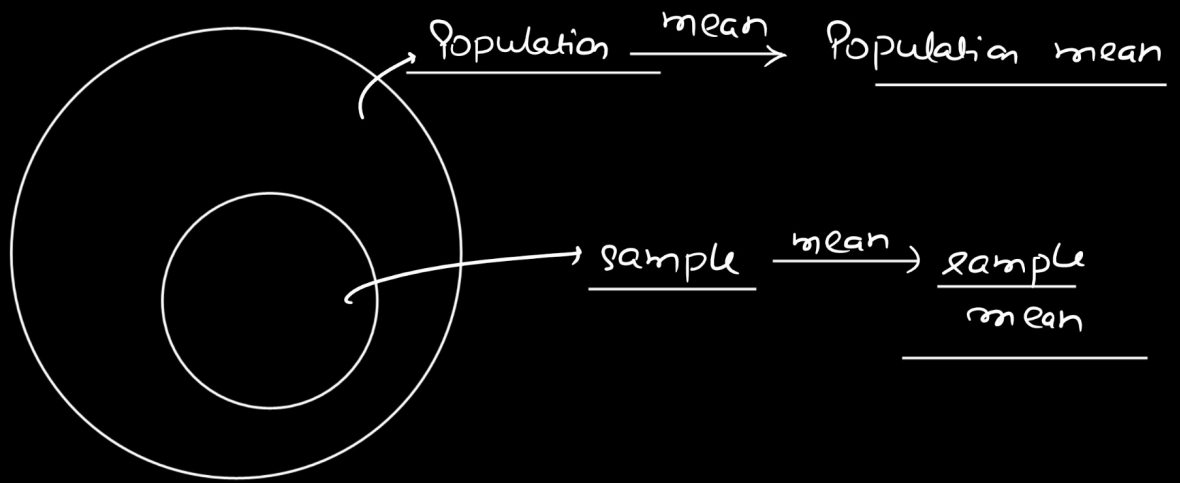
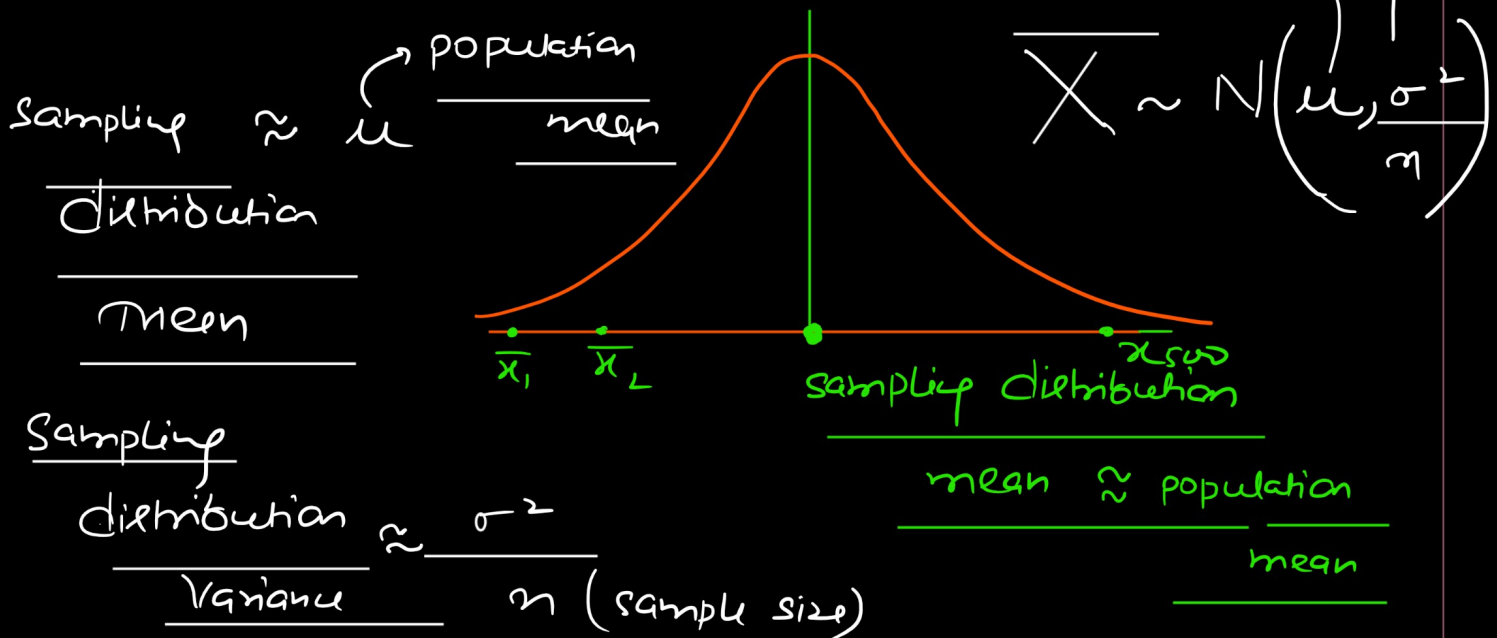
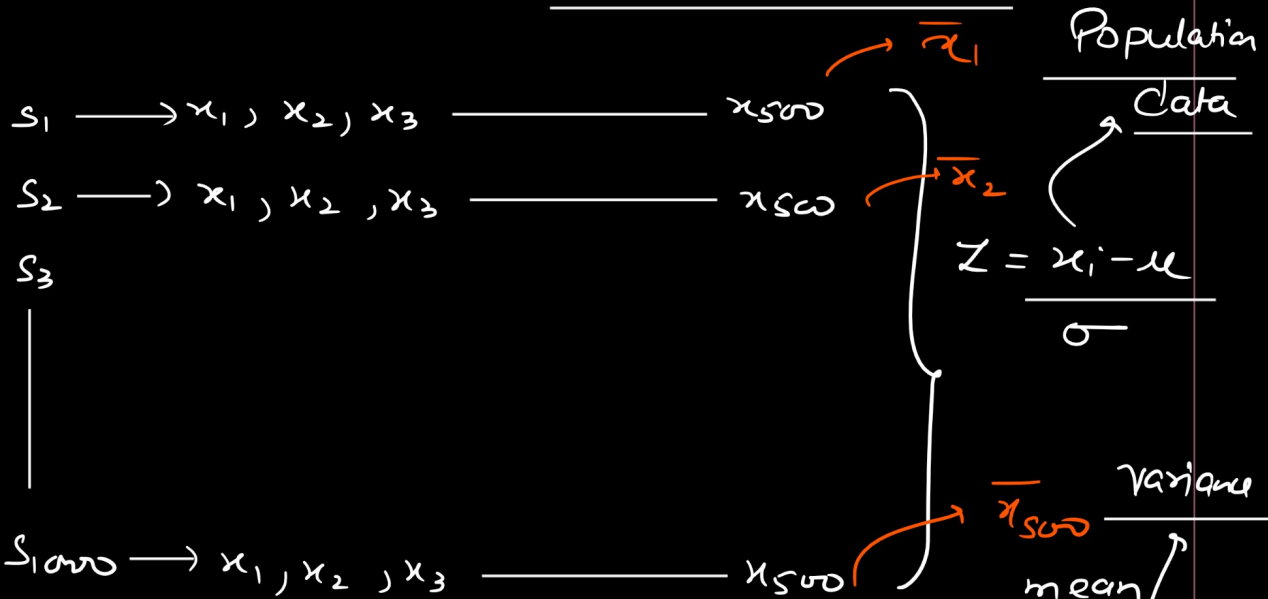


Central Limit Theorem



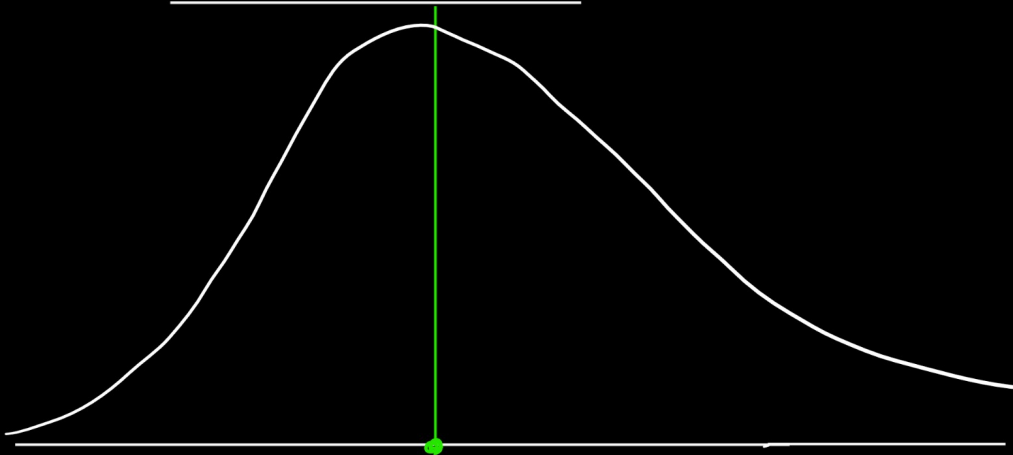
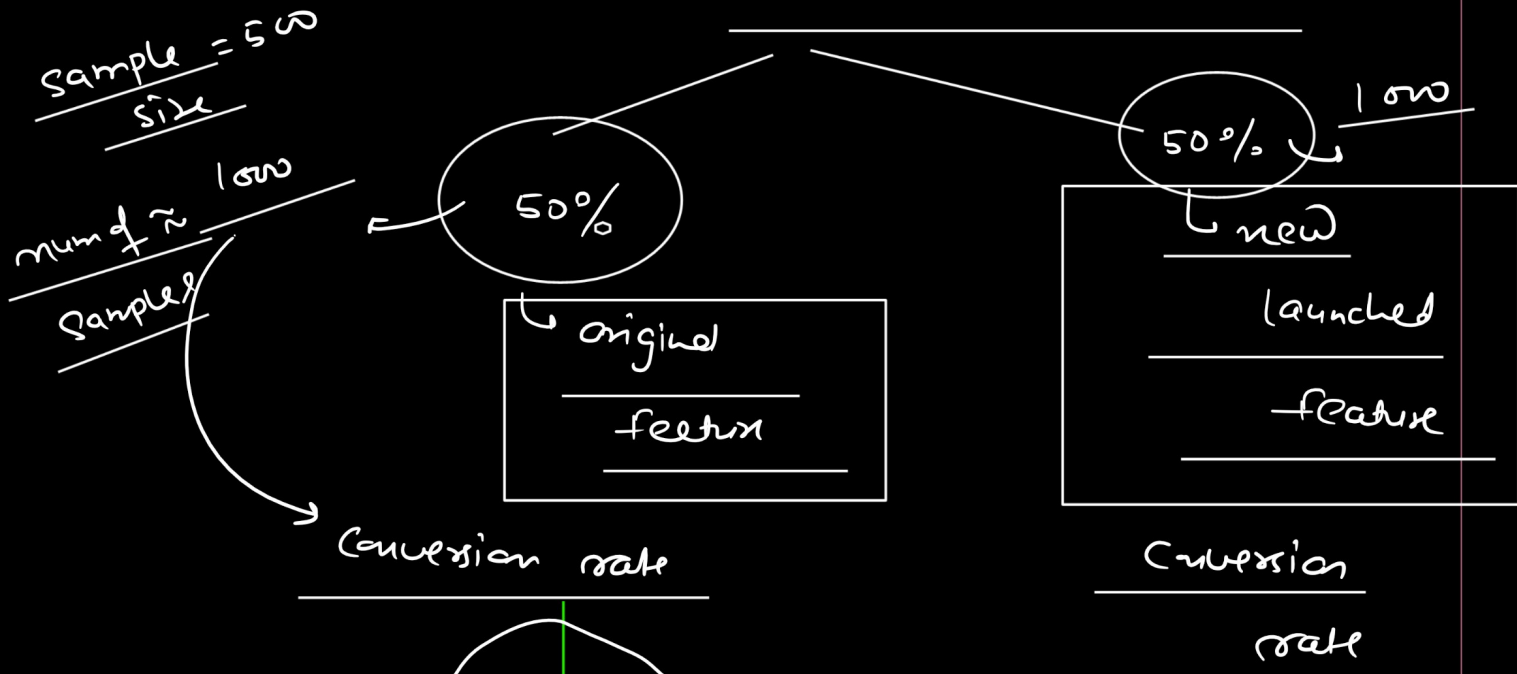
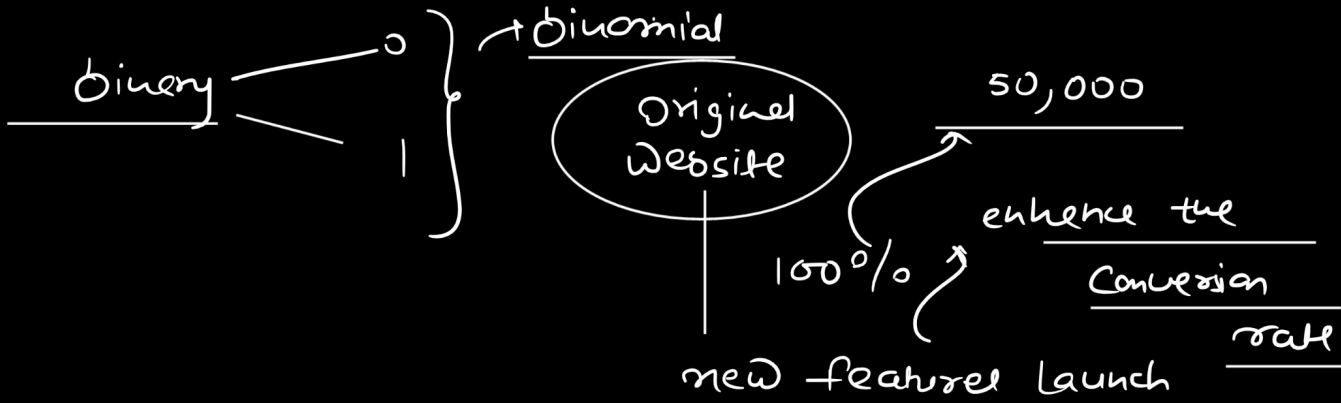
num - of - items = 1000

Sample size = 500



Ideal situation

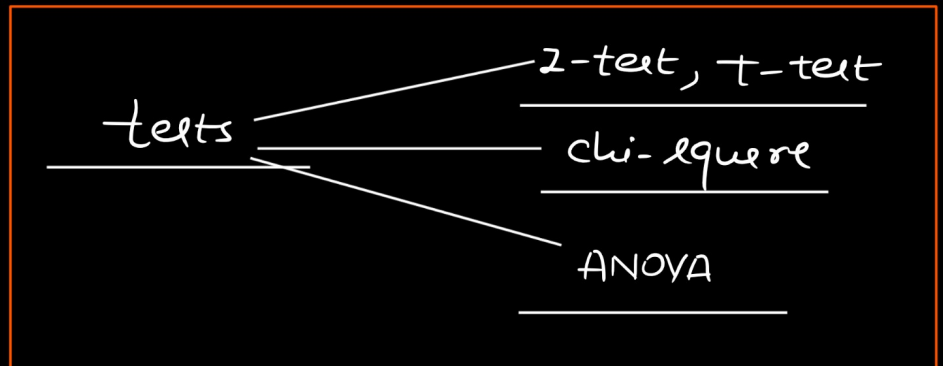
Sample mean $\approx$ population mean



Sampling Distribution mean $\approx$ population mean

structured flow Hypothesis $\rightarrow$ making an assumption

Null Hypothesis $H_0 \rightarrow \mu_{\text{sample}} \approx \mu_{\text{population}}$
Alternate Hypothesis $H_1 \rightarrow \mu_{\text{sample}} \neq \mu_{\text{population}}$
 $\hookrightarrow$ alternate hypothesis



P value $\rightarrow$ probability of H_0 to be true

α $\rightarrow$ significance value (Risk factor)
 $\hookrightarrow$ domain experts

$\alpha \approx 0.05$ (5%) conclude

(Accept H_0)

$p\text{-value} > \alpha \rightarrow$ fail to reject H_0

$p\text{-value} < \alpha \rightarrow$ Reject H_0

z-test / t-test

data is in the form of
Normal Distribution

Population

$$N = 5000$$

$$\mu = \frac{x_1 + x_2 + \dots + x_{5000}}{5000}$$

σ

$$\sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

Sample

$$n = 500$$

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_{500}}{500}$$

s

$$\sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

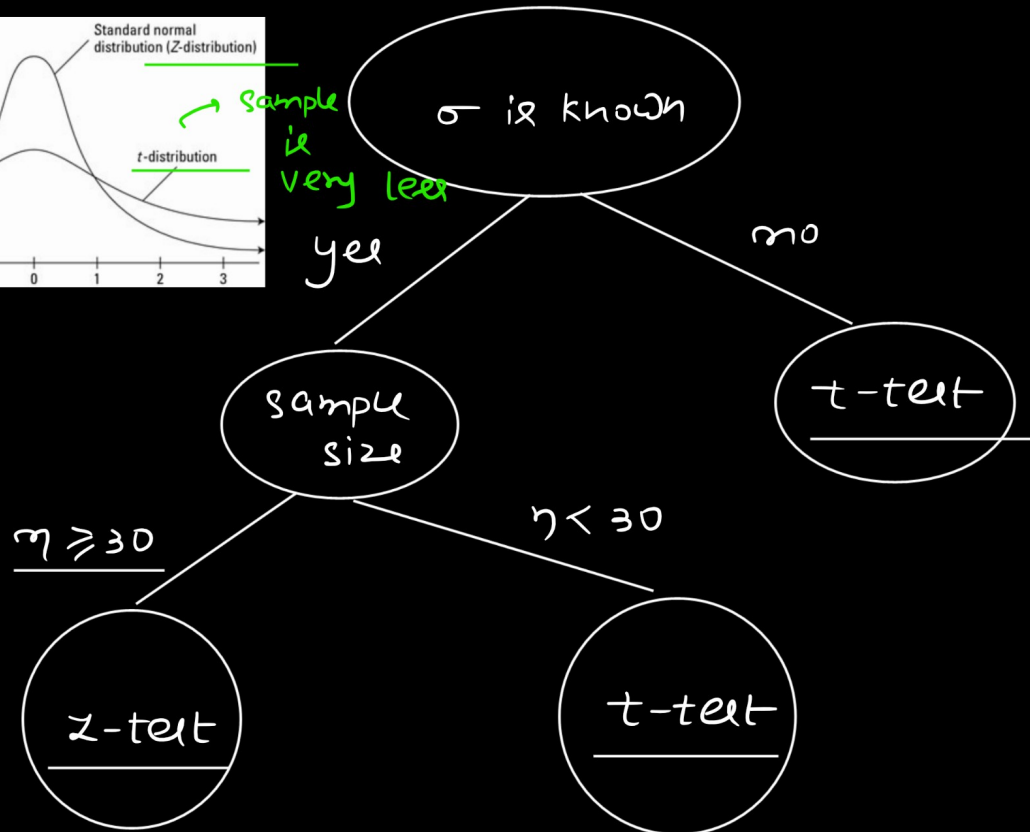
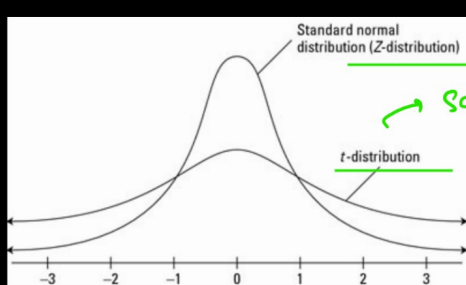
$\approx$

different wrt

population

SD

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$



$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}} \text{ (standard error)}}$$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

A factory claims that the average lifespan of its light bulbs is 800 hours.

To verify this claim, a quality analyst selects a random sample of 50 light bulbs and finds that the sample mean lifespan is 790 hours. The standard error (SE) of the sample is 7.5 hours.

Using a significance level of $\alpha = 0.05$, test whether there is sufficient evidence to conclude that the actual average lifespan of the light bulbs is different from the claimed value.

↗ define the null & alternate hypothesis

$$H_0 \rightarrow \mu = 800$$

$$\mu = 800 \text{ hours}$$

$$H_1 \rightarrow \mu \neq 800$$

$$n = 50$$

$$\bar{x} = 790 \text{ hours}$$

$$SE = 7.5 \text{ hours} \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$\text{Experimental test} \rightarrow \left\{ \begin{array}{l} \text{I-test} \rightarrow \frac{\bar{x} - \mu}{SE} = \frac{790 - 800}{7.5} \end{array} \right.$$

$$\frac{z - 1.33}{}$$

$$\text{P-value} = 0.1835$$

$$\alpha = 0.05$$

$$\text{P-value} < \alpha$$

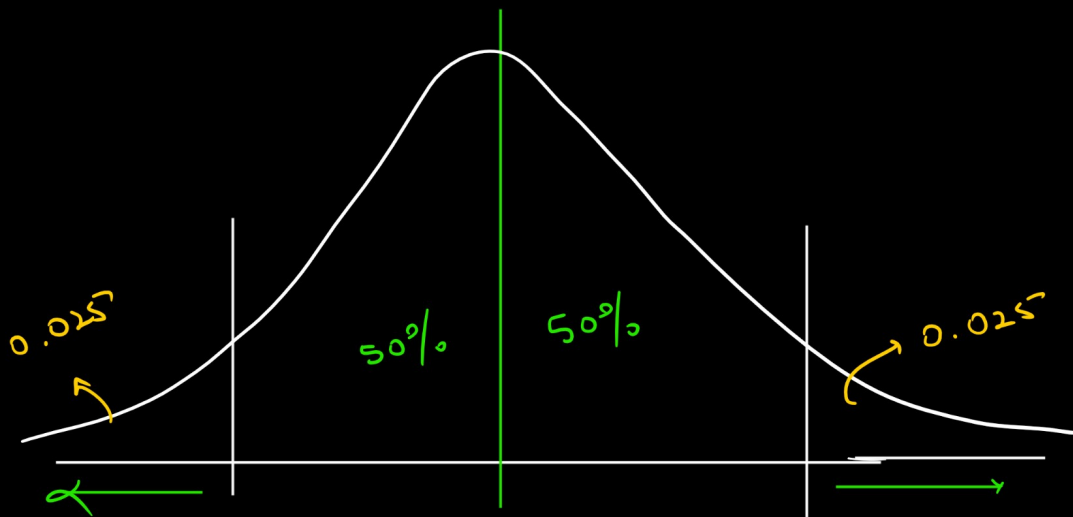
↳ reject H_0

$$\text{p-value} > \alpha \rightarrow \text{fail to reject}$$

H_0



Average lifespan of light bulb is approximately
800 hours only



A nutrition company claims that its protein bars contain an average of 20 grams of protein.

To verify this claim, a sample of 40 protein bars is selected, and the average protein content is found to be 18.8 grams. The standard error (SE) of the sample is 0.8 grams.

Using a significance level of $\alpha = 0.05$, test whether there is sufficient evidence to conclude that the actual average protein content differs from the claimed value.

$$\begin{array}{l}
 \mu = 20 \\
 n = 40 \\
 SE = 0.8 \\
 \bar{x} = 18.8
 \end{array}
 \quad
 \begin{array}{l}
 z = \frac{\bar{x} - \mu}{SE} \\
 z = \frac{18.8 - 20}{0.8} \\
 z = -1.5 \\
 p = 0.1336 > \alpha \\
 \text{fail to reject } H_0
 \end{array}$$

Decision	Reality	
	<u>H_0 is true</u>	<u>H_0 is false</u>
<u>Reject H_0</u>	<u>Type 1 error (α)</u>	<u>Correct ✓</u>
<u>fail to reject H_0</u>	<u>Correct ✓</u>	<u>Type 2 error (β)</u>

An inspector has to choose between certifying a building as safe or saying that the building is not safe.

There are two hypotheses:

(1) Building is safe _____

(2) Building is not safe _____

Here,

H_0 : Building is not safe vs H_a : Building is safe

Type 1 error is committed

When we reject H_0

when H_0 is true in

Type 1 error → H_0 → true

Experiment → Reject H_0

reality

Critical

→ it committed if we reject 'building
is not safe' when it is not

safe.

Type 2 error is committed when we fail to

reject H_0 when H_0 is false

↓

if we fail to reject

'Building is not safe' when it is safe.
